Homework 01 Part 6

Question Prompt:

You are a farmer planting a new crop. You have an empty rectangular field to plant this crop that is 90 ft long and 75 ft wide. Rows of this crop need to be 2 ft wide and 2 ft apart from each other, so you need to account for 4ft total for each row. Within the rows, each seed needs to be planted 1.5 ft away from each other in order to give the plants enough room to establish their roots. All of the rows must be parallel.

Write a Python script that answers the following problems to help you plan your field of crop:

1. How should you configure your rows if you want to maximize the number of seeds you can plant? Will you be able to plant more seeds if you make the rows parallel to the width or parallel to the length? (Remember you can't have part of a row! Only full rows!)
2. With the configuration in your answer to question 1, how many seeds will you be able to plant? Print this number with a label in front of the number saying "Total seeds planted: [INSERT NUMBER]".
3. You need to order seed packets for this crop. The seed packets come with 15 seeds per packet. Answer the following questions in your same Python script, and for each question, print the number in addition to a label that tells you what the number means. (For example, if the answer to question 3a is 37, print a label like "Total packets emptied: 37".)
    a. How many packets will you use and empty? (Hint: there are a couple of ways to find this, but the math module can be useful for this)
    b. How many packets will you need in total? (Hint: the math module can be useful for this)
    c. How many seeds will be left over at the end of planting?
4. In the python script, print an order for the seed packets to send to the supplier that follows the following format:

    Order: [INSERT ANSWER TO QUESTION 3b HERE] seed packets
    Deliver to:
    Purple Farm
    100 Mountain Tree Ln.
    Farmville, MA 00000

5. BONUS: Represent the total number of seeds you will plant in binary.

Explanation of what the question is meant to assess:

The main goal of this question is to assess problem-solving skills by taking a bigger problem and splitting it up into smaller pieces to address with Python. This problem focuses on an overarching question (how many seed packets a farmer needs to fill an empty lot on their farm), and splits it up into smaller pieces (how to configure the rows, how many seeds they can plant in total, how many seed packets they would need). Therefore, this problem will not just assess Python skills, it also allows the person answering the problem to practice their problem-solving abilities. Some of the specific Python skills involved are creating constants, modulo arithmetic, assigning variables, and printing a mulit-line string. Because the question asks the student to print each numerical answer with a label indicating what the number means, the problem assesses the student's ability to add a string to an integer, which must be done by converting the integer to a string. Some modular arithmetic is also assessed. For example, when determining how many rows to have and how many seeds to plant, a farmer cannot have part of a row or part of a seed. Because of this, the script needs to use floor division to determine how many whole rows can fit in the field and how many whole seeds can fit in each row. Furthermore, in question 3c, a good way to answer the question of how many seeds will be left over at the end of planting if each packet has 15 seeds in it is to take the modulo (%) to find the remainder when you divide the total number of seeds needed by the number of seeds per packet. Using this operator is not explicitly stated in the question, but the question causes students to think about the remainder, pointing them towards this operator. Lastly, using the "ceiling" and "floor" functions in the math module are useful in questions 3a and 3b. Because there are a few ways to answer these questions, it assesses the programmer's creativity in addition to their knowledge of the math module.

I think that the most challenging part of this problem to the average student might be figuring out what operations, expressions, and variables to use to answer the questions. I think that once they understand the general idea of the question and how to approach answering it, they will be able to easily apply their Python skills. For me and for many students, understanding each individual skill in Python is relatively simple, but understanding the applications of those skills, how to use them together, when to use them, and how they can solve problems is more challenging. This question, therefore, is intended to help the students overcome the difficulty of knowing which Python skills to use when, and help them learn how to be a better problem-solver.

Answer Key:

One possible answer code to questions 1-4 is found in the file `hw01_part6_answer.py`

Question 5 answer:

